

Chapter 2

Methodology: image analysis, simulation, and DFT-EELS

This chapter provides an introduction to the fundamentals of image analysis and simulation methodologies that are used in conjunction with experimental measurements obtained from transmission electron microscope (TEM), to perform sample characterisation. It begins with the use and applications of image analysis, followed by the theory behind the simulation of Electron Energy Loss Spectroscopy (EELS) using Density Functional Theory (DFT) calculations.

2.1 Image analysis

Scanning Transmission Electron Microscopy (STEM) image analysis is essential for understanding material properties at the atomic scale, providing both qualitative and quantitative insights into structural features. It enables the identification of defects such as vacancies, dislocations, and antisite defects, which play a critical role in determining a material's electronic and mechanical behaviour. Image analysis done on the images obtained from different modes, such as high-angle annular dark field (HAADF) and annular bright field (ABF) (mostly used in this thesis), enables researchers to infer crucial information about atomic structures [1]. Structural quantitative characterisation can be carried out through measurements, such as interplanar spacings [2, 3], strain [4], defect densities [5, 6], and atomic column intensities [7]. This capability is particularly valuable for studying materials with complex microstructures, including nanomaterials and heterostructures. The analysis also helps elucidate how various imaging conditions such as electron dose [8], defocus [9], and aberrations [10] affect the observed contrast and resolution in TEM images.

Modern TEM analysis relies on specialised software platforms that streamline image processing, quantitative analysis, and visualisation. These tools range from proprietary software like Digital Micrograph [11] or Nion Swift [12], Eje-Z [13, 14] to open-source packages such as ATOMAP [2], each tailored to specific aspects of image analysis. With respect to the focus of this thesis, the software ATOMAP has been used extensively for image analysis.

2.1.1 ATOMAP

ATOMAP is an open-source Python-based tool for analysing high-resolution STEM images. It is designed specifically to identify atomic column positions and calculate significant information, which are essential for quantitative studies of atomic arrangements, defects, and strain in materials [2]. ATOMAP combines advanced image processing algorithms and statistical methods, making it highly effective for precise and automated atomic-scale analysis. Some of its most relevant characterisation capabilities are as follows.

- **Atomic Column Detection:** ATOMAP uses Gaussian fitting to locate the centers of atomic columns in STEM images. This technique is based on the assumption that atomic columns appear as Gaussian-shaped intensity peaks in HAADF-STEM images, with the position of each peak corresponding to an atomic column. Gaussian fitting enhances the precision of atomic position determination, as the fitted peak's centroid provides sub-pixel resolution.
- **Atomic Lattice Mapping:** Once the atomic columns are identified, ATOMAP can map the lattice structure by analysing the positions of these atomic columns relative to each other. This lattice mapping step is essential for determining local variations in atomic spacing, which can indicate strain, defects, or compositional variations.

In materials with multiple atomic sub-lattices (such as compounds with different atomic species), ATOMAP can identify and distinguish each sub-lattice. This differentiation allows users to study complex materials where the arrangement of the different sublattices determines the material properties, such as perovskite structures.
- **Atomic Column Intensity Profiling:** In HAADF-STEM images, the intensity of atomic columns is correlated with atomic number (Z-contrast), allowing for compositional related analysis. ATOMAP profiles these intensities to estimate elemental composition or detect changes in occupancy in atomic columns, which is especially useful in studying doped materials or alloys.
- **Ellipticity:** In many HAADF-STEM images, the shape of the columns can also provide important information about the atomic arrangement. Small shifts of the atoms perpendicular to the projection direction result in elongated column shapes. Ellipticity in ATOMAP is calculated by fitting an ellipse to atomic columns from the STEM images. The software calculates the optimum-fit ellipse for sets of atoms to facilitate precise determination of values of ellipticity. Quantitative information about every atomic column in the form of orientation of the major axis of the fitted 2D gaussians and the major to minor axis ratio is also extracted. These values are directly interpretable in the case of HAADF-STEM images as two overlapped atomic columns.

- **Vector Mapping:** From the lattice mapping, ATOMAP calculates vectors between neighbouring atomic columns to analyse distortions like polar off-centering, polarisation gradients, and ferroelectric or antiferroelectric phase transitions.

To illustrate the image analysis performed by ATOMAP, we provide here a brief example, an analysis of a $\text{BiFeO}_3/\text{LaFeO}_3$ (BFO/LFO) multilayer system. The samples were provided by the team of Dr. Pascal Ruello from the Institut des Molécules et Matériaux du Mans (UMR 6283 CNRS), Le Mans Université, and these samples are used for Terahertz (THz) photonics [15] and photovoltaic responses [16]. The BFO/LFO superlattice has been grown with Pulsed Laser Deposition (PLD). In total, 15 bilayers of BFO/LFO have been grown on a strontium titanate substrate for a superlattice thickness of around 60 nm. High Resolution TEM (HRTEM) images and diffraction patterns were obtained in the JEOL 2010F of the CCI-TUB, working at 200 kV and HAADF-STEM images were obtained in a FEI Titan3 G2 in CentraleSupélec (Université Paris-Saclay) working at 300 kV.

Fig. 2.1(a) shows the HRTEM image of the BFO/LFO sample. Figs. 2.1(b) and 2.1(c) show the corresponding Fast Fourier Transforms (FFT) and the intensity profiles along the arrows marked in the FFT, respectively. Fig. 2.1(d) presents the selected area electron diffraction (SAED) pattern, and Fig. 2.1(e) shows the intensity profile along the arrows marked in the SAED pattern, parallel to the growth direction. The preliminary analysis from FFT (Fig. 2.1(b)) reveals the presence of multilayers with two types of periodicity, one with $1/12$ that arises from the succession of 6 unit cells of BFO and 6 of LFO and another periodicity of $1/24$, which would imply a doubling of periodicity. The intensity profile from the SAED pattern (Fig. 2.1(e)) also reveals the same. The origin of this phenomenon couldn't be directly observed in the HRTEM or HAADF-STEM images. This is where ATOMAP was utilised on HAADF-STEM images. At first, ATOMAP is fed the data to locate the various atomic columns (A site and B site in this case) as shown in Fig. 2.2. Once it is done, the ellipticity of the Bi atomic positions (Fig. 2.3(b)) showed a periodic pattern that could be linked to the Bi atoms zig-zagging, resulting in an elliptical atomic column. The ellipticity of the Bi positions is in good agreement with the Pnma antiferroelectric structure proposed

in [17]. Plotting the main axis of the ellipse (Fig. 2.3(a)) shows that the ellipticity pattern is not simply repeated after one BFO/LFO bilayer (which has a period of Λ). Instead, it repeats after two bilayers (2Λ) [15], resulting in a superlattice of 24 unit cells, which explains the electron diffraction and HRTEM FFT observed reflections. This example shows how image analysis is an indispensable tool in the understanding of materials at the atomic level.

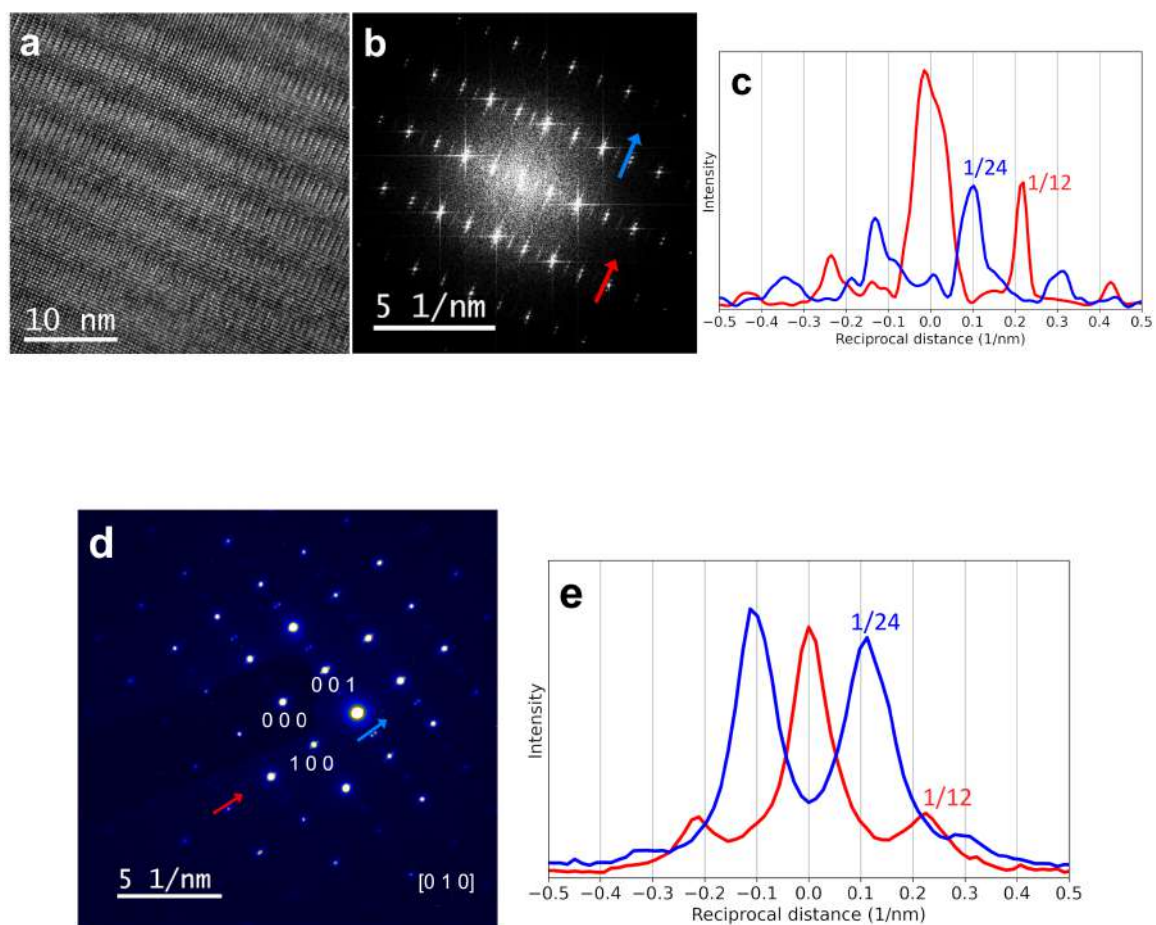


Fig. 2.1 BFO/LFO superlayer depicting the emergence of lattices with different orders of periodicity as per the experimental findings. (a) HRTEM image, (b) FFT, and (c) intensity profile along the corresponding arrows in the FFT. The first satellites of the red profile correspond to 1/12th of the unit cell spot, while the blue peaks are placed at 1/24th of the same reciprocal distance. (d) Diffraction pattern (indexed as pseudocubic symmetry) and (e) the intensity profile along the arrows marked on the diffraction pattern.

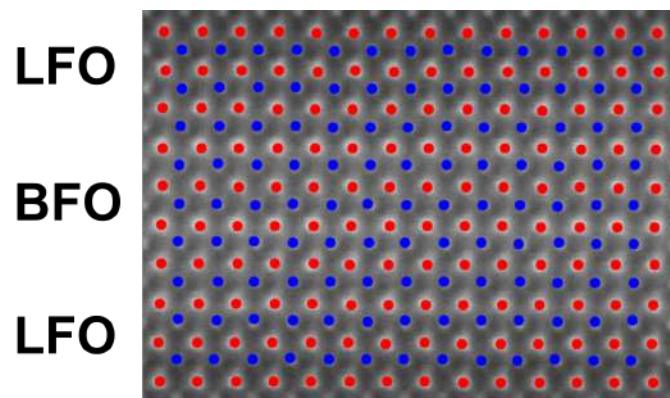


Fig. 2.2 Detection of atomic columns (A site in red, B site in blue) of the BFO/LFO sample using ATOMAP.

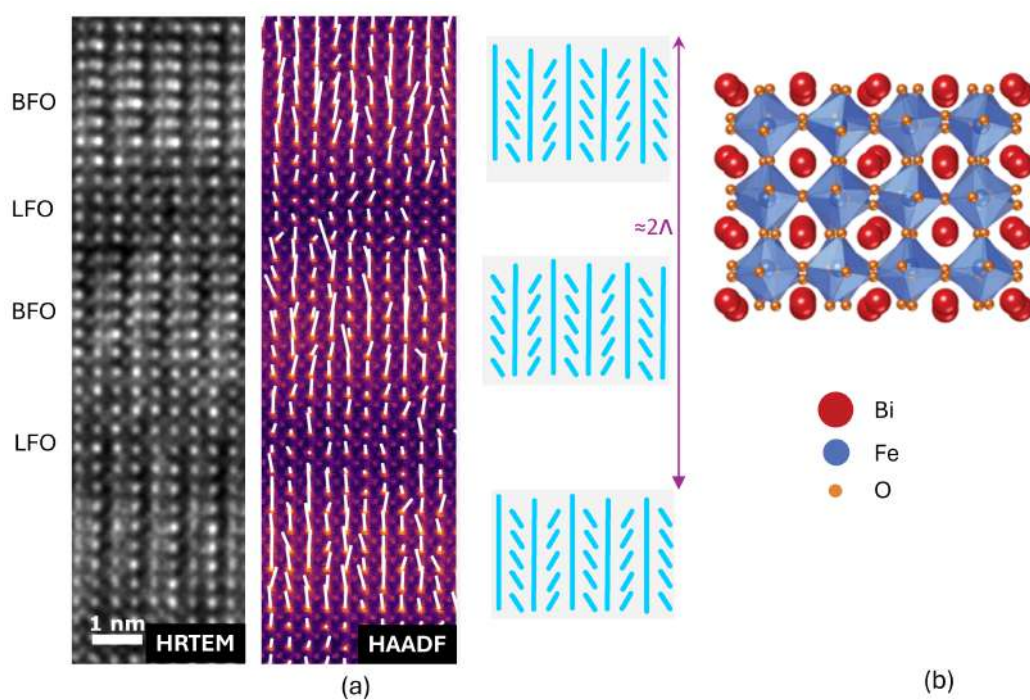


Fig. 2.3 (a) Multiple structural order detection in a BFO/LFO superlattice using ATOMAP. HRTEM image (left) and A-site displacements (by ellipticity in ATOMAP) (middle) extracted from a HAADF-STEM image obtained at 300 kV acceleration voltage. The sketch on the right panel depicts the non-equivalence of the Bi displacement (light blue lines) between two adjacent BFO layers. (b) The crystalline structure of BFO, showcasing the double structural order in the Bi orientations.

2.2 Why do we need image simulation

Interpreting STEM images is often complex due to multiple scattering effects, contrast transfer functions, electron phase interactions, and specimen thickness variations [18]. To overcome these challenges and better understand experimental STEM images, STEM image simulation has become an essential tool in materials science, namely in nanotechnology, functional oxides and semiconductor research. It provides theoretical images based on known structural models, enabling researchers to validate their experimental observations, refine material properties, and interpret complex image features [19].

Simulations help in deconvoluting these effects to extract meaningful structural and compositional information. Simulated images act as a reference, making it easier to distinguish real material features from artefacts introduced by experimental conditions [20]. Another advantage is the verification of atomic structures. By comparing simulated images with experimental ones, researchers can validate structural models of unknown materials, including crystal structures, defects, and interfaces.

2.3 Fundamentals of image simulation

Over the years, STEM image simulation has evolved significantly, with advances in both computational algorithms and hardware, making demanding simulations increasingly accessible. The core principles of STEM image simulation have remained largely the same, but improvements in model accuracy, potential calculation, and electron propagation methods have transformed the field. The two most common methods of STEM image simulations are the Bloch Wave Method [21, 22] and the Multislice Method [23, 24]. The Bloch wave method utilises Bloch waves to describe the propagation of electrons through a periodic crystal lattice. This method is particularly well-suited for small, periodic unit cells and provides highly accurate results for diffraction simulations. The multislice method involves dividing the specimen into thin slices and simulating the electron wave's transmission and propagation through each slice sequentially. It uses fast Fourier transforms to compute the evolution of the electron wave function between slices, making it suitable for large-scale

simulations. The image simulation process typically includes model creation, potential generation, wave propagation, image formation, and post-processing in the order as shown in Fig. 2.4. In the following sections, each of the steps for such a simulation will be explained.

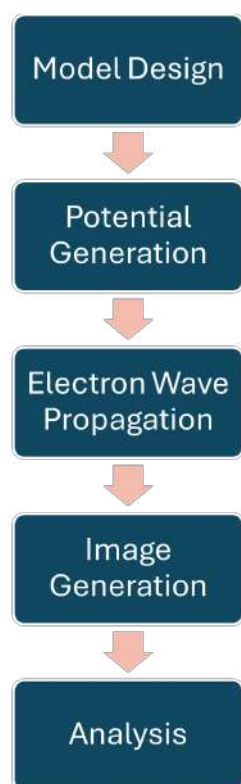


Fig. 2.4 Flow chart depicting the steps for STEM image simulations.

2.3.1 Model design

The model design starts by obtaining the crystallographic data from the crystallographic databases, usually in the form of CIF (Crystallographic Information File). This file provides the lattice parameters and atomic positions within the unit cell, essential for building a periodic model of the crystal. In cases where the crystal data is not available, it can be built manually using multiple softwares such as Vesta [25] and Atomic Simulation Environment [26].

Once the basic structure has been built, depending on the interest of the study, the structure is modified to incorporate changes such as vacancies, defects, dislocations and is used for the next steps. These features affect the local scattering potential and influence

image contrast. In polycrystalline materials or heterostructures, it's important to include grain boundaries or interfaces, as these regions exhibit distinct electronic and structural properties.

Following the introduction of the changes in the structure, structural relaxation is often performed to allow atoms to find their lowest energy configuration, which minimises any artificial stress introduced during defect incorporation. This is usually done by performing Density Functional Theory (DFT) calculations. These methods refine the atomic positions, to provide a realistic model that accurately simulates electron scattering effects.

2.3.2 Potential generation

Potential generation involves defining how the atomic structure of the sample interacts with the incident electron wave by calculating the scattering potential at each atomic site. The primary purpose of potential generation is to produce a static map of scattering potentials that align with the atomic positions in the sample. This scattering potential represents the electrostatic field produced by the atomic nuclei and surrounding electron cloud of each atom. The resulting potential map forms the basis for simulating electron scattering within the material. Although this map remains constant throughout the simulation, it underpins the interaction between the electron wave and the sample. The multislice approach of generating potentials is one of the most used methods for such tasks. In this method, the phase shift for each slice is calculated separately, based on the atomic potentials within that slice. This approach is critical in thick samples where dynamical scattering effects become significant, as it models the cumulative effect of electron scattering through each layer.

2.3.3 Electron wave propagation (multislice simulation)

Electron wave propagation is a critical step in the process of multislice STEM image simulation. It is the stage where the electron wave interacts dynamically with the sample atomic structure, based on the scattering potentials defined in earlier steps. The multislice method captures dynamical scattering effects [27] and interference patterns that arise from the electron wave interacting with atoms at various depths, which is essential for accurately simulating high-resolution STEM images.

Traditional approximations, like the kinematic theory, assume only single scattering events and therefore are insufficient for high atomic number materials or thicker samples where multiple scattering events are significant. In the multislice approach, the sample is divided into a series of thin, parallel slices, each representing a specific atomic layer or a group of layers as shown in Fig. 2.5. The electron wave is modified as it propagates through each slice, capturing the complex scattering effects that arise from the material's structure.

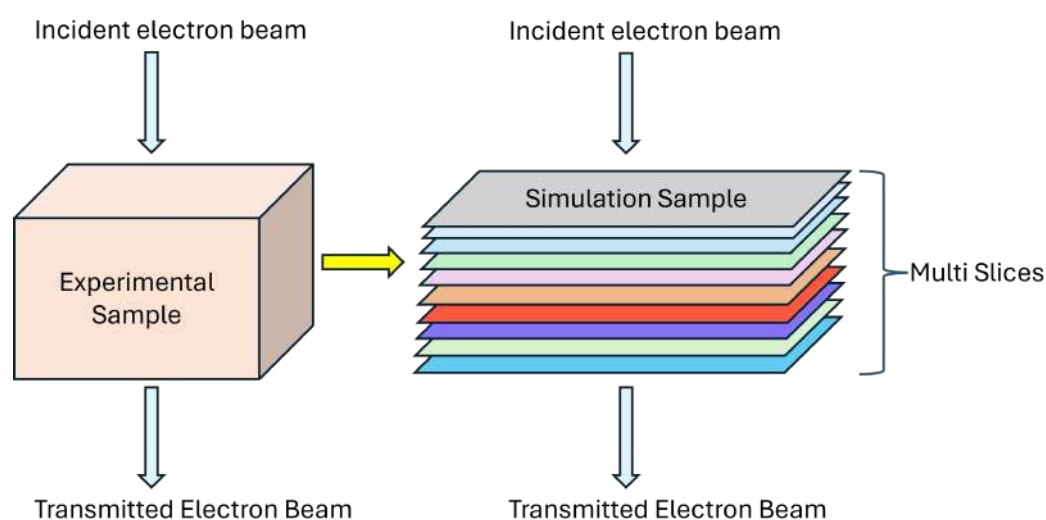


Fig. 2.5 Conversion of an experimental sample into its equivalent multislice sample for simulation.

The initial electron wave function, $\psi_0(r)$, represents the coherent, plane electron wave entering the sample. This wave function is defined based on parameters such as the electron accelerating voltage and incident beam coherence. For example, a typical accelerating voltage for high-resolution STEM is around 300 kV, which corresponds to an electron wavelength of approximately $0.0197 \text{ \AA}$, enabling resolutions sufficient to resolve atomic spacings in most crystalline materials [20].

At each slice, the electron wave undergoes a phase shift due to the electrostatic potential generated by the atomic structure within that layer. This phase shift is governed by the projected potential of the atoms in the slice, denoted as $V(r)$, and is calculated based on the

atomic scattering potentials previously generated. The phase shift imparted by each slice is described by the transmission function $T(r)$:

$$T(r) = \exp\left(i\sigma \frac{V(r)}{\lambda}\right),$$

where σ is the interaction constant (dependent on the electron wavelength and accelerating voltage) and λ is the electron wavelength. The function $T(r)$ applies a phase modulation to the wave function according to the electrostatic potential of each slice, effectively representing the impact of atomic structure on the electron wave [19].

After interacting with the potential in a slice, the electron wave is propagated to the next slice using a free-space propagation function. This propagation accounts for the distance Δz between slices and involves calculating how the electron wave diffracts over that distance. In Fourier space, this propagation is represented by a propagation operator $P(k)$, which applies a phase factor corresponding to the free-space travel distance:

$$P(k) = \exp\left(-i\pi \frac{\lambda \Delta z}{k}\right),$$

where k is the spatial frequency vector in Fourier space. The propagation operator effectively translates the electron wave from one atomic layer to the next, with phase alterations that model diffraction effects over the slice thickness. This operation is repeated for each slice, building up the cumulative effect of electron scattering through the entire thickness of the sample [27].

The propagation and phase-shifting process is iterated over all slices, building up a cumulative interaction between the electron wave and the sample. At each step, the wave function $\psi_n(r)$ is transformed by applying the phase shift from the transmission function $T(r)$ and then propagated using the operator $P(k)$. This iterative process captures dynamical scattering effects, where the electron wave repeatedly scatters within the material, leading

to complex interference patterns and contrast features in the final image. Such dynamical effects are particularly important for thick samples and materials with heavy atoms, such as transition metal oxides [19]. The resulting exit wave function is then used in the image generation process.

2.3.4 Image generation

Once the electron wave exits the sample, the simulated image formation step models how the microscope's optical elements, such as the objective lens and aperture, transform the electron wave into an intensity distribution that corresponds to the observed STEM image.

The objective lens in a STEM introduces various aberrations, including defocus, astigmatism, coma, spherical and chromatic aberrations, which distort the electron wave. Correcting for these aberrations is crucial for high-resolution imaging and is simulated using objective lens transfer functions. The contrast observed in STEM images is also influenced by phase shifts introduced by the said lens. All of these must be taken into account in this step.

Beyond the sample, several detectors such as bright field (BF), high angle annular dark field (HAADF), annular dark/bright field detectors (ADF/ABF) or even segmented detectors can be simulated by choosing the angular range of electrons that will form the simulated image. HAADF and ABF modes are particularly useful in distinguishing between atoms of different atomic masses, with HAADF favoring heavier elements and ABF allowing the visualisation of lighter atoms.

For all the image simulations in this thesis, the software Dr. Probe [24] has been used. It is based on the multislice algorithm and features the frozen-lattice method for simulating thermal diffuse scattering by averaging over many random atomic displacements, which is critical for accurate atom counting and bias prevention in high-resolution simulations. The various settings used for the simulations are explained in the relevant chapter. To illustrate the use of detectors and image simulations, an example is included in the following paragraph.

$\text{YBa}_2\text{Cu}_3\text{O}_7$ is a high-temperature superconducting oxide with a layered perovskite structure, widely studied for its exceptional electronic properties [28–30]. Its complex atomic arrangement, consisting of alternating layers of heavy (Ba, Cu, Y) and light (O) elements

as shown in Fig. 2.6(a), makes it suitable for evaluating contrast mechanisms and imaging capabilities in STEM. An example of STEM image simulation of $\text{YBa}_2\text{Cu}_3\text{O}_7$ is presented in Fig. 2.6(b,c). In this study, a crystallographic supercell was first constructed to represent the atomic arrangement, and STEM image simulations were subsequently performed. The simulated images clearly demonstrate the contrast sensitivity of different detectors: in the HAADF mode as shown in Fig. 2.6(b), only the heavier elements (Ba, Cu, Y) are distinctly visible due to the Z-contrast mechanism, whereas in the ABF image as shown in Fig. 2.6(c), the lighter oxygen atoms are also resolved, showcasing the technique's capability to capture low-Z elements. The iterative nature of the multislice method enables the realistic simulation of STEM images by accurately capturing the structure and composition of the sample. This approach is particularly effective for materials with complex atomic structures or high atomic numbers, as it accounts for cumulative phase modulation across layers. This is crucial for interpreting lattice strain, atomic positions, and defects, providing insights into the material's magnetic and electronic properties, thereby making this method an essential tool for STEM image simulation [18]. Such simulations not only validate the theoretical expectations but also serve as a benchmark for interpreting experimental STEM data. They contribute significantly to our understanding of the imaging process and structural identification in complex oxides, especially in the context of functional materials like superconductors.

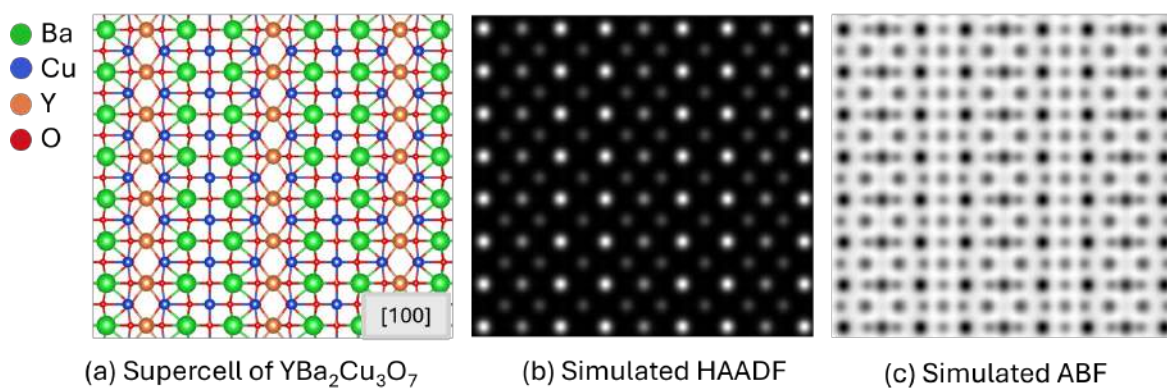


Fig. 2.6 Simulation of STEM imaging of (a) supercell of $\text{YBa}_2\text{Cu}_3\text{O}_7$, its corresponding HAADF simulation (b) and ABF simulation (c) .

2.4 Why do we need DFT

The study of nanomaterials, particularly crystalline and polycrystalline materials typically involves the study of several atoms, usually periodically arranged in a particular order referred as a quantum system. The theoretical study of these quantum systems is a complex field as they stand as a many body problem having multiple properties such as dual wave/particle like behaviour, existence in multiple states simultaneously, time dependent change of states or, interaction with the electronic charges of electron beams. Unlike classical mechanics, which can predict the motion of macroscopic objects with certainty, quantum mechanics deals with the inherently probabilistic nature of subatomic particles. This introduces unique challenges in characterising the behaviour of particles at the quantum level. In order to solve them, the Schrödinger equation provides a mathematical framework for understanding and predicting the behaviour of quantum systems in diverse contexts [31]. The equation addresses the particle duality through wave functions, which encapsulate the probabilities of finding a particle in various states. It allows for the description of a system's evolution over time, considering the contributions of all possible states. It predicts discrete energy states, as observed in atoms and molecules, thus providing a theoretical basis for the understanding of atomic structure and spectral lines. Thus, its solutions yield critical insights into the nature of matter at the atomic and molecular levels [32].

The Schrödinger Equation can be written in two forms (time dependent and time independent). The time dependent equation describes the evolution of a quantum system over time. It is given by

$$i\hbar \frac{\partial}{\partial t} \Psi(\mathbf{r}, t) = \hat{H} \Psi(\mathbf{r}, t) \quad (2.1)$$

Where:

- $\Psi(\mathbf{r}, t)$ is the wavefunction, a complex function that contains all the information about the quantum state of the system.
- $\hbar$ is the reduced Planck's constant, defined as $\hbar = \frac{h}{2\pi}$.

- $\hat{H}$ is the Hamiltonian operator, representing the total energy of the system, including both kinetic and potential energies.

In case the potential energy does not depend on time, the equation changes to a time-independent one and is given by

$$\hat{H}\Psi(\mathbf{r}) = E\Psi(\mathbf{r}) \quad (2.2)$$

Where:

- E is the total energy eigenvalue associated with the wavefunction $\Psi(\mathbf{r})$.

Schrödinger equation can easily be solved for a system of few atoms but in case of a many-body system, there are multiple interacting electrons which increases the number of variables as well as complicates the Hamiltonians. Further exchange effects become significant in such systems and the potential energy spread is a combination of multiple types of interactions (ionic, covalent, Van der Waals, etc.), which makes the equations highly non linear and difficult to solve. Moreover, complex materials often exist in non ideal conditions subjected to external electrical or magnetic fields and also possess disorders. These factors introduce additional complications in accurately solving the Schrödinger equation. The computational resources required for solving the Schrödinger equation for complex systems scale exponentially with the number of electrons. This necessitates the development of advanced computational methods and approximations to analyse and predict the behaviour of complex materials, in short DFT.

2.5 Core principles of DFT

DFT focuses on the electron density $\rho(r)$ rather than on the many-electron wavefunction Ψ (Fig. 2.7). This simplifies the problem since $\rho(r)$ depends only on three spatial variables, regardless of the number of electrons in the system. Effectively, it includes the average effects of electron-electron interactions, which is often sufficient for many practical applications.

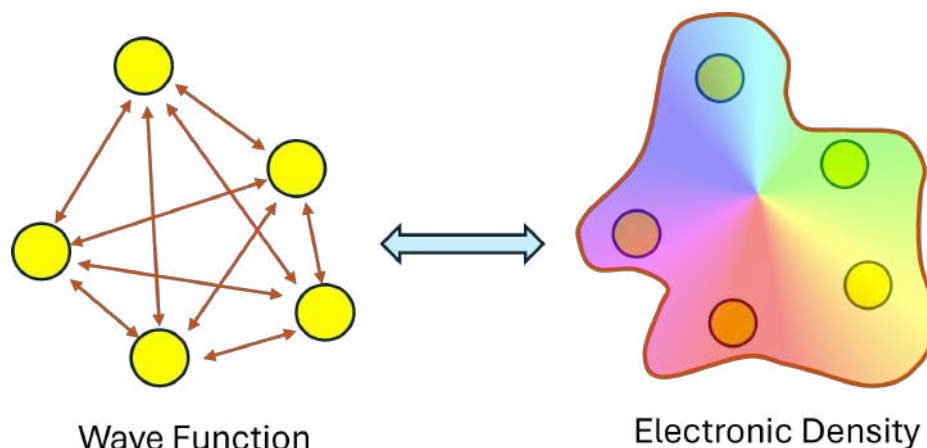


Fig. 2.7 Simplification of Schrödinger equation from wave functions to electronic densities in DFT.

The Hohenberg-Kohn (HK) theorems, established in 1964 by Pierre Hohenberg and Walter Kohn, are the foundation of Density Functional Theory (DFT) [33, 34]. These theorems shifted the focus from the many-body wavefunction to the electron density. The theorems are as follows:

- **First theorem:** The external potential (and hence the total energy) of a many-electron system is a unique functional of the electron density $\rho(\mathbf{r})$. This implies that knowing the electron density is equivalent to knowing the full wavefunction.
- **Second theorem:** The ground-state energy can be obtained by minimising the functional of the electron density, $E[\rho]$, where the true ground-state density gives the minimum energy.

These theorems demonstrate that the ground-state properties of a many-electron system are uniquely determined by an electron density that depends on only three spatial coordinates, greatly simplifying the many-body problem.

The Kohn-Sham (KS) equations rely on the idea that the exact ground-state electron density $\rho(\mathbf{r})$ of a system of interacting electrons can be represented by the electron density of a system of non-interacting electrons moving in an effective potential $V_{\text{eff}}(\mathbf{r})$. The KS equations are a set of self-consistent equations that describe this non-interacting system. For

each electron, the KS equation takes the form of a one-electron Schrödinger-like equation:

$$\left(-\frac{\hbar^2}{2m} \nabla^2 + V_{\text{eff}}(\mathbf{r}) \right) \psi_i(\mathbf{r}) = \varepsilon_i \psi_i(\mathbf{r}), \quad (2.3)$$

where:

- $\psi_i(\mathbf{r})$ are the KS orbitals (one-electron wavefunctions).
- ε_i are the orbital energies.
- $V_{\text{eff}}(\mathbf{r})$ is the effective potential, defined as:

$$V_{\text{eff}}(\mathbf{r}) = V_{\text{ext}}(\mathbf{r}) + V_H(\mathbf{r}) + V_{\text{xc}}(\mathbf{r}).$$

- **External Potential** $V_{\text{ext}}(\mathbf{r})$: This is the potential due to atomic nuclei or any other external field acting on the electrons.
- **Hartree Potential** $V_H(\mathbf{r})$: This term describes the classical electrostatic interaction between electrons. It is calculated from the electron density:

$$V_H(\mathbf{r}) = \int \frac{\rho(\mathbf{r}')}{|\mathbf{r} - \mathbf{r}'|} d^3r'.$$

The Hartree potential accounts for the repulsion between electrons, treating them as classical point charges.

- **Exchange-Correlation Potential** $V_{\text{xc}}(\mathbf{r})$: It includes the exchange effects arising from the Pauli exclusion principle, where electrons with the same spin avoid each other and the correlation effects which arise from the correlated motion of electrons due to their mutual repulsion. It is approximated using models like the Local Density Approximation (LDA), Generalised Gradient Approximation (GGA) or Perdew–Burke–Ernzerhof (PBE) functional.

The expression of the total exchange-correlation can be expressed as

$$E_{\text{xc}}[\rho] = E_{\text{x}}[\rho] + E_{\text{c}}[\rho]$$

Where $E_x[\rho]$ is the exchange energy functional corresponding to the energy transfer between electrons of the same spin and $E_c[\rho]$ corresponds to the correlation energy between opposite spin electrons.

2.6 Flow of steps in DFT logic calculation

Using exchange-correlation (XC) functionals and treating the system as non-interacting, one can solve the KS equations. The resulting solutions, referred to as KS orbitals, must be obtained in a self-consistent manner. The KS orbitals, electron densities, and the Hamiltonian are interrelated, and all these elements must be considered to achieve convergence.

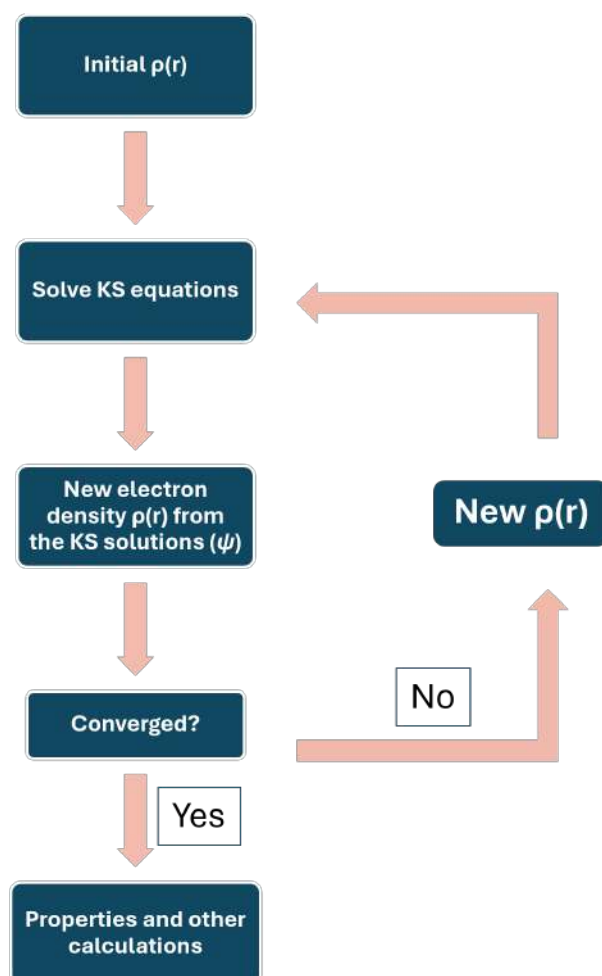


Fig. 2.8 Scheme representing the flow of DFT algorithm.

The convergence scheme has been shown in Fig. 2.8. The algorithm begins by choosing an initial guess for the electron density $\rho(r)$. The electron density is used to solve the KS equations iteratively to find the wavefunction ψ . Using this new ψ , the $\rho(r)$ is updated. The updated $\rho(r)$ is compared to the previous $\rho(r)$. If the difference between the new and old densities is below a pre-defined threshold, the solution is considered converged. If convergence has not been achieved, a new density is generated by mixing the old and new densities, a method known as density mixing. The algorithm then returns to step 2 and the cycle repeats till the convergence is met. Once the convergence is achieved, the required properties and other calculations can be performed as needed.

2.7 EELS and ELNES from DFT

The calculation of EELS and ELNES spectra using DFT typically involves three main steps:

1. **Density of States (DOS) calculation:** The first step in the process involves calculating the DOS by solving the KS equations within DFT. This yields the energy levels (eigenvalues) and electronic wave functions, which are then used to construct the DOS. The total DOS represents the number of available electronic states at each energy level, while the partial DOS (PDOS) provides insights into the orbital and atomic contributions.
2. **Optical properties calculation:** Using the DOS, DFT calculates optical properties such as the dielectric function, $\epsilon(\omega)$, which describes the material's response to electric fields. The imaginary part of the dielectric function, derived from interband transitions, is particularly important as it represents the material's absorptive properties and is foundational in modelling energy loss.
3. **EELS and ELNES spectrum generation:** The final step utilises the dielectric function and DOS to simulate the EELS and ELNES spectra. In low loss EELS, the energy loss function, $\text{Im}\left(-\frac{1}{\epsilon(\omega)}\right)$, is derived from the dielectric function, capturing plasmonic and

interband excitations. For ELNES, transitions from core levels to unoccupied states are calculated from PDOS features, reflecting the local bonding and electronic structure in the near-edge region.

2.7.1 DOS and PDOS calculation

The eigenvalues ϵ_i represent the energies of the electronic states, and the wave functions $\psi_i(\mathbf{r})$ describe the distribution of electrons. In DFT, we can obtain this by solving the KS equation as mentioned in equation 2.3.

The density of states (DOS) is calculated from the eigenvalues ϵ_i by summing over all energy levels:

$$D(E) = \sum_i \delta(E - \epsilon_i), \quad (2.4)$$

where $D(E)$ is the total DOS at energy E , and $\delta(E - \epsilon_i)$ is the Dirac delta function, which counts the states at each eigenvalue. In practical implementations, a small broadening factor (usually Gaussian or Lorentzian) is applied to smooth the DOS curve, making it easier to interpret.

For PDOS, which provides insights into the contributions of specific atomic orbitals or elements, the DOS is projected onto these specific atomic states (e.g., s , p , and d -orbitals). The PDOS can reveal orbital-specific features that are essential in interpreting bonding and spectral characteristics [35].

2.7.2 Optical properties calculation

Once the DOS is obtained, the optical properties of the material can be calculated. These properties are foundational for simulating EELS spectra, as EELS measures the energy lost by electrons due to interactions with the electronic structure of the material.

The dielectric function,

$$\epsilon(\omega) = \epsilon_1(\omega) + i\epsilon_2(\omega),$$

describes the material's response to an oscillating electric field. For EELS, the imaginary part $\epsilon_2(\omega)$ is particularly important, as it represents the absorption within the material and reflects the material's electronic structure.

The imaginary part of the dielectric function is given by:

$$\epsilon_2(\omega) = \frac{8\pi^2 e^2}{\Omega} \sum_k \sum_{i,j} \frac{|\langle \psi_j(k) | \mathbf{e} \cdot \mathbf{r} | \psi_i(k) \rangle|^2}{(E_j(k) - E_i(k))^2} \delta(\hbar\omega - (E_j(k) - E_i(k))),$$

where:

- $\mathbf{e}$ is the polarization vector,
- $E_i(k)$ and $E_j(k)$ are the initial and final state energies,
- $\delta(\hbar\omega - (E_j(k) - E_i(k)))$ represents the energy conservation condition, ensuring transitions occur only at the specified photon energy $\hbar\omega$ [36].

This equation considers all possible transitions between occupied and unoccupied states, weighted by the transition matrix elements and the DOS at each energy. The real part $\epsilon_1(\omega)$ of the dielectric function can be calculated from $\epsilon_2(\omega)$ using the Kramers-Kronig relations, which link the real and imaginary components.

The energy loss function (ELF) for simulating low loss EELS spectra, is given by:

$$\text{Im} \left(-\frac{1}{\epsilon(\omega)} \right) = \frac{\epsilon_2^2(\omega)}{\epsilon_1^2(\omega) + \epsilon_2^2(\omega)} \text{Im}(-\epsilon(\omega))$$

This function represents the energy lost by electrons as they traverse the material and undergo excitations. Peaks in this function indicate plasmon resonances and interband transitions, providing insights into both the bulk and surface plasmonic properties of the material [37].

2.7.3 EELS and ELNES spectrum generation

With the DOS and optical properties calculated, the final step is generating the EELS and ELNES spectra. EELS focuses on the collective and single-electron excitations within

the material, while ELNES examines fine structure near ionisation edges, reflecting local bonding and coordination environments.

ELNES focuses on the fine structure observed in the energy-loss spectrum. It is sensitive to the local electronic structure and the chemical environment of specific atoms within a material. In ELNES, electrons are excited from core levels (e.g., 1s, 2p) to unoccupied states above the Fermi level. The characteristics of these transitions depend heavily on the local bonding environment and the nature of the unoccupied states. Its spectrum exhibits sharp features, often described as peaks or shoulders, which correspond to specific electronic transitions. These features are highly sensitive to the symmetry and hybridisation of the bonding orbitals, providing detailed information about the local coordination of atoms. The intensity of the ELNES spectrum can be expressed as:

$$I(\omega) = \sum_{i,f} |\langle \psi_f | \mathbf{q} \cdot \mathbf{r} | \psi_i \rangle|^2 \delta(E_f - E_i - \hbar\omega)$$

where ψ_i and ψ_f are the initial and final state wave functions, $\mathbf{q}$ is the momentum transfer vector, and $\delta(E_f - E_i - \hbar\omega)$ ensures energy conservation for the transitions. The spectral features depend on the local density of states (LDOS) near the Fermi level, highlighting how local environments affect electronic transitions.

To illustrate the use of the simulations to interpret experimental changes in ELNES, we provide an analysis for the identification of lithiated and delithiated regions in lithium iron phosphate (LiFePO_4 , henceforward referred as LFP) cathodes from a solid oxide fuel cell, using both Li K edge and Fe $L_{2,3}$ and $M_{2,3}$ edges. The samples were provided by the team of Dr. Alejandro Morata Garcia from the Nanoionics and Fuel Cells department of the Catalonia Institute for Energy Research (IREC). These materials are typically used in solid state battery industries [38]. The bilayer structure consisting of LFP and $\text{Li}_{1.3}\text{Al}_{0.3}\text{Ti}_{1.7}(\text{PO}_4)_3$ (LATP) thin films was deposited on silicon chips coated with platinum. Following deposition, the samples underwent rapid thermal processing (RTP) at 800°C for 60 seconds under an argon atmosphere with a ramp rate of 10°C/s to achieve crystallinity and interfacial adhesion.

In the primary analysis of LFP cathodes, by applying clustering methods [39] on the experimental EELS data as obtained from the spectrum image shown in Fig. 2.9(a), two

distinct regions emerged, labelled as bulk and defect regions, as shown in Fig. 2.9(b). These two regions presented differences in both, low loss (Fig. 2.10(c)) and high energy loss spectra (Fig. 2.9(c)). A hypothesis was made that the defective regions might remain delithiated after several charge cycles. To account for the presence and/or absence of Li, ELNES of LiFePO_4 and FePO_4 were simulated as shown in Fig. 2.9(d). In LiFePO_4 and FePO_4 , Fe has an oxidation state of Fe(II) and Fe(III) respectively. Upon observing the fine structure of Fe (by comparing Fig. 2.9(c) and 2.9(d)), it was possible to note distinct features that can be linked to the simulated ELNES. The bulk region, with maxima at lower energies and thinner peaks correspond well to the simulated LFP with Fe(II). Conversely, the defective region is similar to Fe(III) owing to maxima appearing at higher energies and having thicker peaks. This points towards a fully lithiated bulk LFP and delithiated defects.

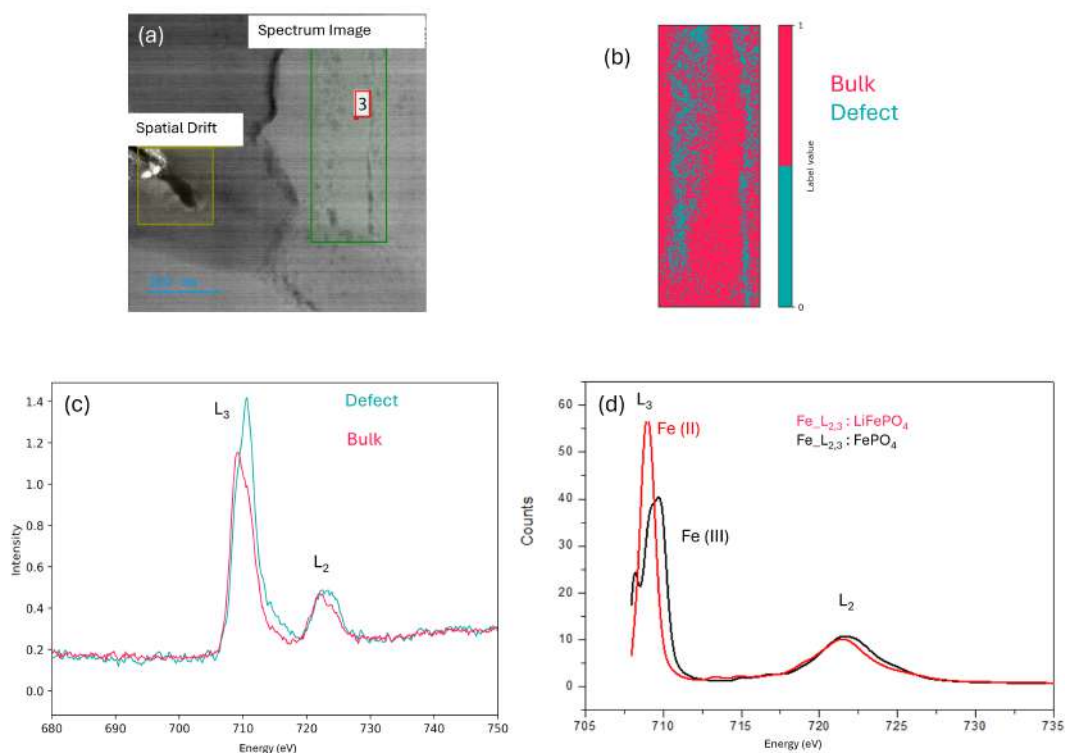


Fig. 2.9 Comparison of Fe $L_{2,3}$ in bulk and defective regions of LFP cathodes using ELNES simulations. (a) EELS spectrum image (b) Experimental EELS region, with areas of the bulk and defect, (c) Experimental EELS averaged spectrum, (d) Simulated ELNES of Fe of two different phases.

Low loss EELS experimental and simulation results are shown in Fig. 2.10. In the low loss region, a small bump can be observed after the Fe $M_{2,3}$ edge only present in the areas without defects (bulk) (Fig. 2.10(c)). This is not clearly observed in the Fe $M_{2,3}$ simulations (Fig. 2.10(d)) but corresponds well to the simulated Li K edge energy (Fig. 2.10(e)) (note the small intensity of the Li transition when compared with the Fe). This confirms that the defective area also lacks Li which is in good correspondence to the change in Fe oxidation state observed in the core-loss region.

Therefore, ELNES simulation has been the key in understanding the experimental data. By employing these techniques in tandem with other DFT calculations, we can aim to achieve a comprehensive understanding of material properties at the atomic scale.

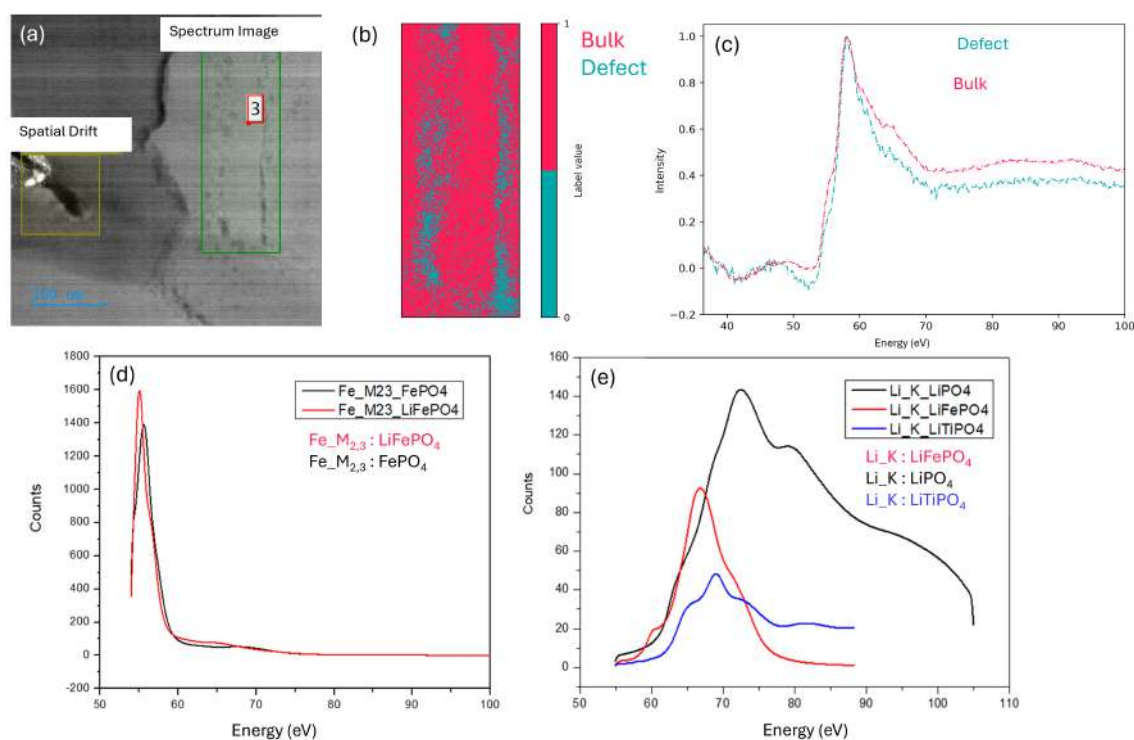


Fig. 2.10 Low loss EELS comparison of bulk and defect regions of LFP cathodes using ELNES simulations (a) EELS spectrum image (b) Experimental EELS region, with areas of the bulk and defect, (c) Experimental EELS averaged spectrum, (d) Simulated $M_{2,3}$ Fe edge of two different phases, (e) Simulated low loss EELS of Li K edge in three different phases.

2.8 References

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